An AI-Driven Intent-Based Network Architecture

Yosra Njah, Aris Leivadeas, and Matthias Falkner

Abstract

Next-generation communication systems are envisioned to offer fully autonomous networks, where infrastructure components will demonstrate self-X capabilities. The emerging paradigm of intent-based networking (IBN) facilitates autonomous operations and user-friendly management, eliminating the need for continuous human intervention. This article offers a comprehensive overview of the IBN paradigm, its core technologies, and its transformative impact on network resilience and efficiency, proposing an end-to-end artificial intelligence (AI)-powered IBN architecture (IBNA). The article initially explores the main components of an IBNA, which encompasses structured iterations of refinement, activation, and assurance. Then, it presents network softwarization and AI-powered schemes for each component. Finally, it consolidates all the proposed schemes into one comprehensive pipeline, demonstrating the feasibility of a full IBNA implementation.

Introduction

In the era of smart environments rapid technological advancements and widespread device connectivity significantly enhance quotidian events. These digital ecosystems continue to experience massive expansion, spanning highly dynamic networks and connecting billions of devices. This rapid proliferation of connected devices, coupled with the increasing demand for heterogeneous applications, has intensified the complexity of network management. Traditionally, network management has been human-driven, which can be error-prone and inadequate to meet the increasing demands of modern networks. Consequently, network operators are facing a significant challenge in efficiently managing highly dynamic, heterogeneous, and extensive infrastructures.

Autonomic networking enables network systems to achieve self-X properties, where X can represent all management capabilities, including configuration, optimization, healing, and so on. The first step toward autonomous networking is currently in progress through network softwarization, leveraging technologies like Network Function Virtualization (NFV) and Software-Defined Networking (SDN), which enhances network and service management but also optimizes agility and efficiency across networks. However, despite significant strides in softwarization, especially with the integration of Artificial Intelligence (AI), manual efforts persist, and the risk of human errors remains a challenge.

To achieve a fully autonomous or zero-touch network, the emerging paradigm of Intent-Based Networking (IBN) takes center stage in the realm of network softwarization. IBN enables users to express their desired network outcomes using human-readable "intents." The system then autonomously refines (translates) these high-level intents into low-level machine-readable policy configurations. Each policy is then set for activation in the infrastructure. At the same time, a crucial component of IBN is intent assurance, which advances network automation by guaranteeing the deployment of generated policy configurations. The continuous verification in this IBN closed-loop automation, which includes refinement, activation, and assurance, secures the alignment with the user intent, eliminating the need for human intervention.

In this article, we introduce the main components of an AI-driven IBN architecture (IBNA), enabling network management with cognitive capabilities. Specifically, in the next section, we explore background notions of the IBN paradigm. Following that, the intent refinement method is first explored relying on Large Language Models (LLM). Then, for the intent activation, recent trends of Deep Reinforcement Learning (DRL) are investigated to address conflicts among intents and with the network state. Additionally, for the intent assurance, a multi-agent DRL (MADRL) approach is explored, ensuring the deployment of heterogeneous intents. The article continues with a comprehensive pipeline integrating the proposed schemes, through a healthcare 4.0 use case. We then present open research challenges and future directions, and finally, we conclude the article.

IBN: Essentials and Advances

The main objective of IBN is to realize autonomous networks, enabling experts and network operators to manage them without constant supervision. Hence, the entire IBN paradigm is undergoing extensive research and standardization. For instance, various networking standards developing organizations (SDOs), such as the Internet Engineering Task Force (IETF) [1], the 3rd Generation Partnership Project (3GPP) [2], and the Open Networking Foundation (ONF) [3], as well as open-source projects like CAMARA [4], actively contribute to defining and standardizing the IBN paradigm. Moreover, IBN has recently

Digital Object Identifier: 10.1109/MCOM.001.2400143

Yosra Njah and Aris Leivadeas are with École de technologie supérieure, Canada; Matthias Falkner is with Cisco Systems Inc., Canada.

0163-6804/25/$25.00 © 2025 IEEE

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.

attracted increasing attention from the research community, resulting in the publication of scientific surveys [5]. However, no single organization provides a full IBNA; instead, comprehensive solutions emerge from collaborative efforts between SDOs and the wider research community. In the subsequent part of the section, we will define the specifics of each IBN component, as shown in Fig. 1, while offering concise insights into relevant existing contributions. Additionally, Table 1 visually describes these components, delineating SDOs' and academic contributions.

Intent refinement is the first phase of an IBNA, where the high-level intent is translated into a network policy. This process varies based on the type of intent expression used, such as a technical network parameter or natural language input. A technical network parameter approach provides interfaces like command line, RESTful, or GUIs, allowing network users to express their intent with specific network parameters. While this approach facilitates the intent translation, it limits eligible users to network administrators, [2, 3]. On the other hand, natural language processing is evolving. Techniques like Amazon voice-assisted technologies with Latent Dirichlet Allocation (LDA) [6], Bidirectional Long Short-Term Memory (LSTM) [7], and Bidirectional Encoder Representations from Transformers (BERT) [15] have been proposed as intent refinement schemes.

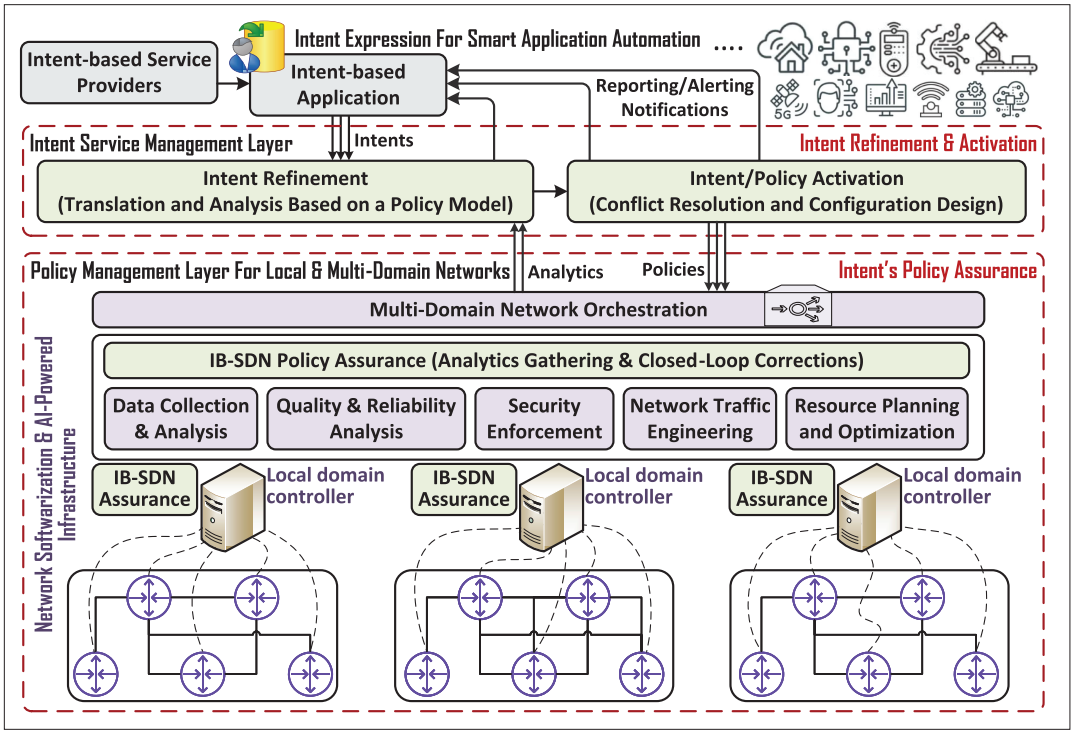


FIGURE 1. High-level workflow of IBN automation.

IBN Components		Sources	Brief Description and Limitations
Mono-Components	Intent Refinement	IETF [1]	Defines intent profiling based on users, types, network scopes, abstraction levels, and life cycles.
		3GPP [2]	GUI-based management system interpreting intents and converting them into specific operations tasks.
		[6]	NLP/LDA-based approach, which may not be suitable for small datasets with many topics.
		[7]	NLP/DL-based system (LSTM-CRF), which lacks details about dataset, use cases, and network policy models.
	Intent Activation	ONF [3]	Advances SDN by incorporating IBN northbound interfaces.
		[8–9]	Traditional formal logic-based systems with filtering and prioritization rules for inter-policy conflict resolution.
		[10]	Traditional ACL updates aiming to resolve conflicts between intent policies and network state configurations.
	Intent Assurance	CAMARA [4]	IBN APIs for managing QoS across diverse networks, facilitating flexible network service provision.
		[11]	Data dissemination system for vehicular edge computing, lacking integration of broader AI techniques.
		[12]	AI-driven QoS control approach, lacking scalability to handle diverse network functionalities.
Full IBN Architecture		SDOs	No SDO provides a full IBNA; holistic solutions need combined efforts from SDOs and the research community.
		[13]	IBNA presenting ambiguity concerning the inter-policy conflict resolution and intent assurance subsystems.
		[14]	IBNA lacking AI integration, restricting automation and adaptability.

TABLE 1. A Summary analysis of IBN initiatives: Examples of SDOs' efforts and relevant research solutions.

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.

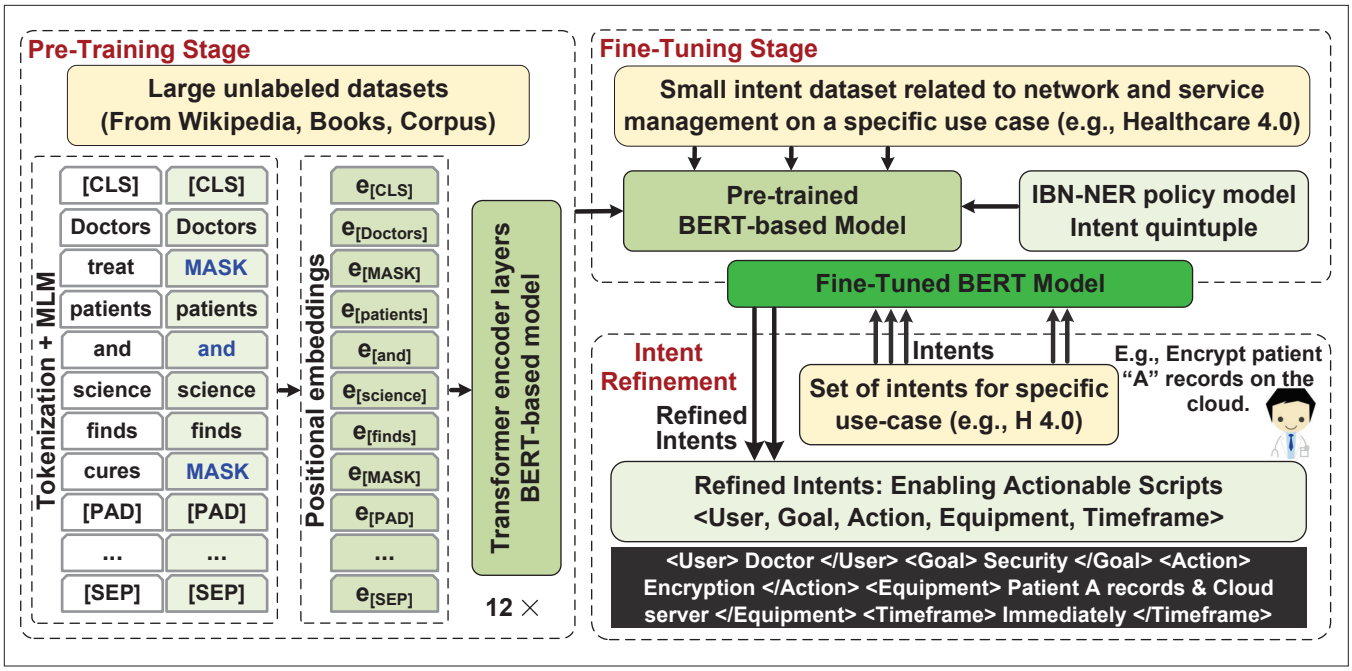


FIGURE 2. Intent refinement using a LLM like BERT.

Intent activation automatically deploys the refined high-level intents in the network infrastructure. Nonetheless, before arriving to the actual deployment, intent conflicts may arise, leading to violations and inconsistencies during network configuration. To ensure conflict-free policy configurations, intent activation should perform two sub-steps: detecting and resolving inter-policy conflicts, and conducting pre-configuration profiling to address conflicts between service requirements and network state. In the current state of IBN research, this component is receiving relatively minimal prominence. Traditionally, policies have been represented as graphs, but current resolution approaches are limited to only Access Control List (ACL) conflict resolutions [10]. In terms of IBN, the few existing works have primarily addressed conflict resolution by employing formal logic with filtering and prioritization rules, focusing on basic connectivity intents [8], or across multiple intent scopes [9].

Intent assurance assures the performance of already deployed intents. As shown in Fig. 1, it is tightly coupled with SDN controllers and orchestration, largely based on telemetry tools. Thus, by continuously monitoring feedback from network functionalities, intent assurance enables an IBN system to meet the intent's requirements throughout its lifetime. In addition to intra-domain or local domain network (e.g., enterprise) that could be managed by a single controller, intents may spread across multi-domain networks (e.g., 5/6G). In these cases, the SDN orchestrators and controllers play a key role in coordinating assurance by connecting to each other and sharing information from the different domains. This IBN component has received considerable attention in the literature within various smart environments, particularly in the context of the Internet of Connected Vehicles [11], addressing specific network functionalities, such as QoS provisioning [12].

An AI-Driven IBNA

Intent Refinement

In our proposed IBNA, the intent is expressed in the form of natural language. The goal is to translate a high-level intent into a network policy model that later on and through a refinement process can be more easily mapped into low-level network configuration scripts. This network policy model should reflect the network capabilities in a granular fashion. For instance, a simplistic network policy model can comprise of a 5-tuple of entities ⟨*user*, *application goal*, *network action*, *target equipment*, *timeframe*⟩.The "user" entity can incorporate the intent definers, including both technical and non-technical users. The "application goal" entity can support users' demands and objectives in terms of network services and applications. The "network action" could specify the network service requirements to fulfill the user's intent. The "target equipment" represents the infrastructure for deploying the user's intent. Lastly, the "timeframe" defines the period that the required service is scheduled to occur. It is important to display all entities within a user's utterance; otherwise, a default setup will be applied for each non-filled entity. For example, within a Hospital 4.0 context, if a doctor asks, "*Encrypt patient A's records on the cloud server*," a possible entity identification would be: {*user: Doctor*}, {*goal: Security*}, {*network action: Encryption*}, {*Equipment: Patient A's records & Cloud server*}, {*Immediately (default configuration)*}. Obviously, the network policy module should be easily extendable and customizable to align the intent requirements to a business-specific domain.

To perform this policy entity identification, an LLM such as BERT is employed, which typically involves two main phases: pre-training and fine-tuning, as depicted in Fig. 2. At first, word tokenization, padding, and integration of position-

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.

al embeddings are applied to each text segment (e.g., intent) before being processed by a set of encoders. Each encoder includes a multi-head self-attention layer for relationship determination and a feedforward neural network for enhanced representation. Following, the system generates contextualized output embeddings, where the same token will have different output embeddings depending on its surrounding terms. These embeddings are enhanced from the Masked Language Model (MLM), a technique used during pre-training to predict masked tokens. Specifically, the MLM addresses the "see itself" issue by predicting masked tokens, enhancing thus the model's contextual comprehension. BERT, like LLMs in general, relies on extensive corpora during the pre-training stage, but can be later fine-tuned for an IBN setting, using a labeled intent dataset. Thus, fine-tuning capacity for specific domains is a notable innovation, allowing parameter adjustments for customized use cases without the need to initiate training anew. This dual approach underscores the versatility and efficacy of LLMs in natural language processing, motivating its inclusion in our IBNA.

Intent-Policy Activation

During IBN policy activation, conflicts may arise in two main forms: inter-policy conflicts, and conflicts between intent requirements and network state. For the first type of conflicts, an inherent part of their resolution is the priority scheme followed. As a rule of thumb, policies coming from network administrators have the highest priority, policies coming from a group of users have medium priority, and policies coming from individual users have the lowest priority. We start by defining key conflict terms, inspired by [8] and [9], to assist readers in understanding subsequent resolution methods used for the first type of conflicts.

Inter-Policy Conflict Detection and Resolution: Conflicts arise when two or more intents aim to access the same network services and resources, typically falling into the following categories, as depicted at the top of Fig. 3:

- *Redundancy* occurs when multiple policies duplicate management operations. As an example, consider policy P1 ⟨Employee 1 and 2 of Group A, High security traffic exchange, Firewall & Encryption, PC of Employee 1 and 2, anytime⟩ and policy P2 ⟨Group A, High security traffic exchange, Firewall & Encryption, PCs of Group A, anytime⟩. Given that employees 1 and 2 belong to group A, the first policy is redundant and should not be activated.
- *Shadowing* occurs when one policy supersedes another by encompassing all operations, rendering the shadowed one inactive. For instance, considering policy P2 ⟨Employee 1 of Group A, Low security traffic exchange, Firewall, PC of Employee 1, anytime⟩ and P1 ⟨Group A, High security traffic exchange, Firewall & Encryption, all PCs of Group A, anytime⟩, policy P2 is shadowed by P1, since P2 is a subset of P1 and requests a contradicting configuration. Additionally, since P1 has a higher priority, P2 should not be enforced.

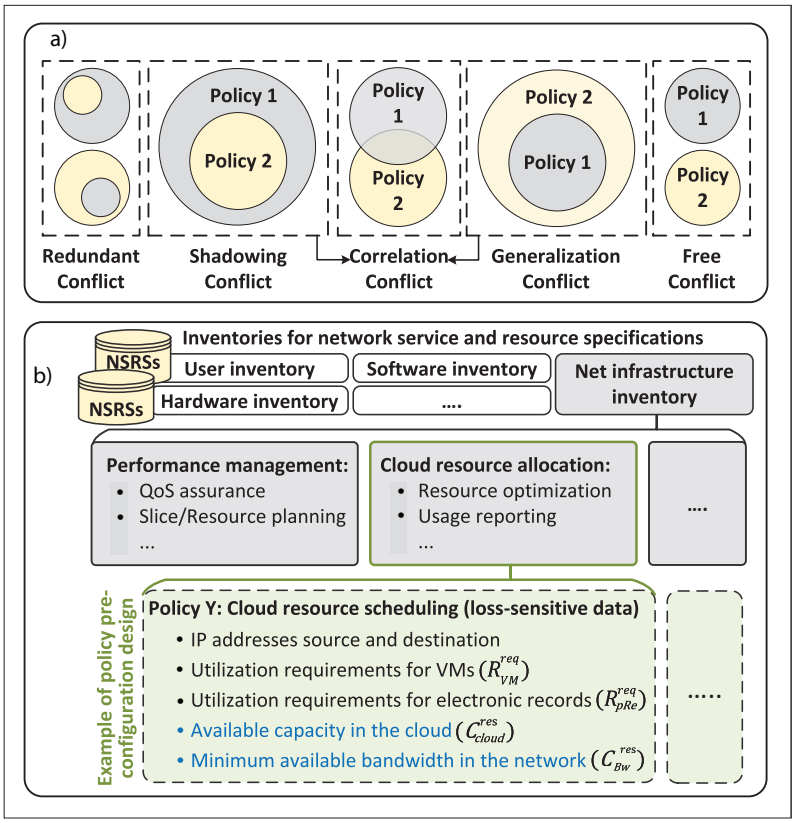


FIGURE 3. Conflict identification and pre-configuration analysis for intent activation: a) Inter-policy conflicts; b) Planning inventory and policy pre-configuration design.

- *Correlation* occurs when two policies partially cover the same operations. As an example, a policy P1 ⟨Employee 1 & 2 of Group A, default QoS videoconferencing, best effort routing, PC of Employee 1 and 2, anytime⟩ is correlated with policy P2 ⟨Employee 2 & 3 of Group A, high QoS videoconferencing, high QoS routing, PC of Employee 2 and 3, anytime⟩, since the employee 2 creates a contradicting intersection. A resolution scenario is to increase the QoS and routing level of P1 to high to lift the contradiction and satisfy both policies.
- *Generalization* occurs when a policy supersedes existing policies with broader operations. Let us assume policy P1 ⟨Admins, high security & default QoS, Firewall & Encryption & best effort routing, all equipment, anytime⟩, and policy P2 ⟨Security Admins, high security & high QoS, Firewall & Encryption & high QoS routing, all equipment, 2029/09/23; 5-11 pm⟩. In this case, P1 is a superset of P2 (since security administrators belong to the administrators group), but P2 has a higher priority, due to a scheduled security upgrade. To resolve such conflicts, policy P1 should exclude the security administrators from the user space during the upgrade.

Policy resolution can be benefit by a DRL framework, empowering an AI agent to detect and resolve inter-policy conflicts. Using sequential decision-making techniques driven by environmental feedback, the DRL framework can dynamically adjust policy spaces and integrate conflict-free policies into the system. For instance, upon detecting redundancy, the DRL agent can

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.

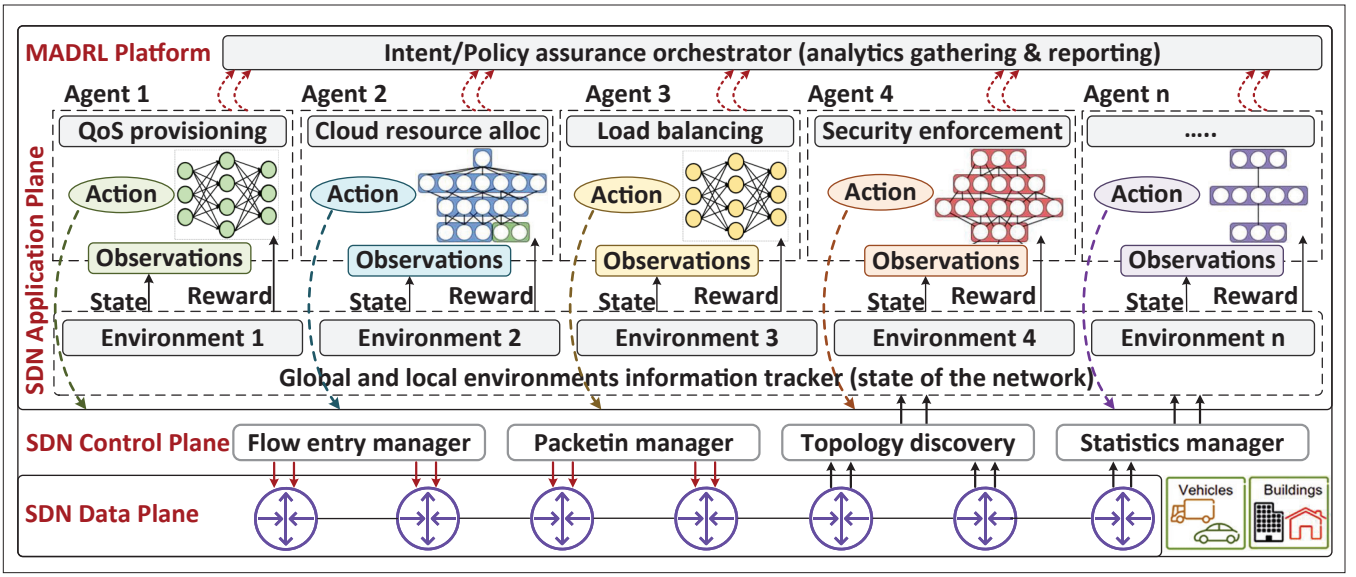


FIGURE 4. A MADRL framework for policy assurance.

eliminate redundant policies. Similarly, with shadowing, it adjusts policy scopes to remove overlaps. In cases of correlation conflicts, adjustments can be made to one policy's space by removing shared elements. In cases of generalization, the agent can deactivate overly broad policies. Common spaces between conflicting policies could be subsequently removed from the generalized policy. The reward mechanism, by also considering the priority scheme, can drive the conflict resolution, ensuring the generation of a conflict-free policy sets. Evidently, as DRL is a sub-optimal solution, in some cases a 100 percent conflict-free solution may not be possible. In case the conflicts cannot be resolved, and as shown in Fig. 1, reporting and alerting notifications will be triggered for the network administrators in order to find a remedy solution.

Conflicts Between Intents and Network State: These types of conflicts emerge if the intents request non-existent services or the requirements exceed the current network capacity. To detect such types of conflicts, a pre-configuration consistency analysis should be performed that can rely on inventories (service specifications) that will verify the feasibility of an intent deployment. These inventories will transfer the actual network state to the policy activation component and combine data on users, services, and network facilities in order to compile configuration script templates detailing the resource specifications and configurations of the translated intents as shown at the bottom of Fig. 3. These inventories will collectively serve as a baseline for evaluating various concurrent policies competing for resources, identifying inconsistencies, and detecting conflicts if the inventories cannot meet the requirements of multiple policies.

Similarly to inter-policy conflict management, a DRL-based system can be extended for resolving conflicts between multiple intents and the network state. Within this framework, the RL agent can deploy action and reward spaces using a prioritization approach to validate feasible high-priority policies and maximize the number of intents that can coexist over the same infrastructure.

Intent-Policy Assurance

For the IBN assurance, our design features a MADRL framework that ensures the effectiveness of heterogeneous network policies, as illustrated in Fig. 4. In this case, network functionalities for policy assurance (e.g., QoS provisioning, cloud resource allocation, security enforcement, etc.) are represented by diverse distributed agents. These agents interact with an observable softwarized network infrastructure, making decisions through independent and cooperative strategies. Each DRL-agent performs actions tailored to optimize its specific objectives. Concurrently, all agents collaborate to achieve efficient network management, regulated by mechanisms that promote cooperation and alignment with overall network goals.

Consider a scenario where a set of heterogeneous, conflict-free, and atomic intents require assurance through various network functionalities. For instance, an intent to ensure high QoS for a video streaming service must coexist with intents for robust security and efficient cloud resource allocation. Each translated intent will be assigned to specific network functionalities (engines). Employing a Markov Decision Process (MDP) approach enables to design distinct state spaces, action spaces, and reward functions for each agent. The global MADRL system aims to maximize the overall reward function over time, such as optimizing resource utilization and minimizing wastage, while each agent independently assures its associated intent requirements.

In more details, each agent autonomously selects actions based on its network objectives, learning from and interacting with its local environment, provided by the SDN controller during infrastructure statistics collection rounds. While agents do not directly communicate during decision-making, they indirectly influence each other through shared local environments. In the meantime, the reward functions align the agents' actions with both individual and overall network optimization objectives by rewarding agents for meeting service fulfillment, while penalizing selfish behavior. This approach balances local agent goals with

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.

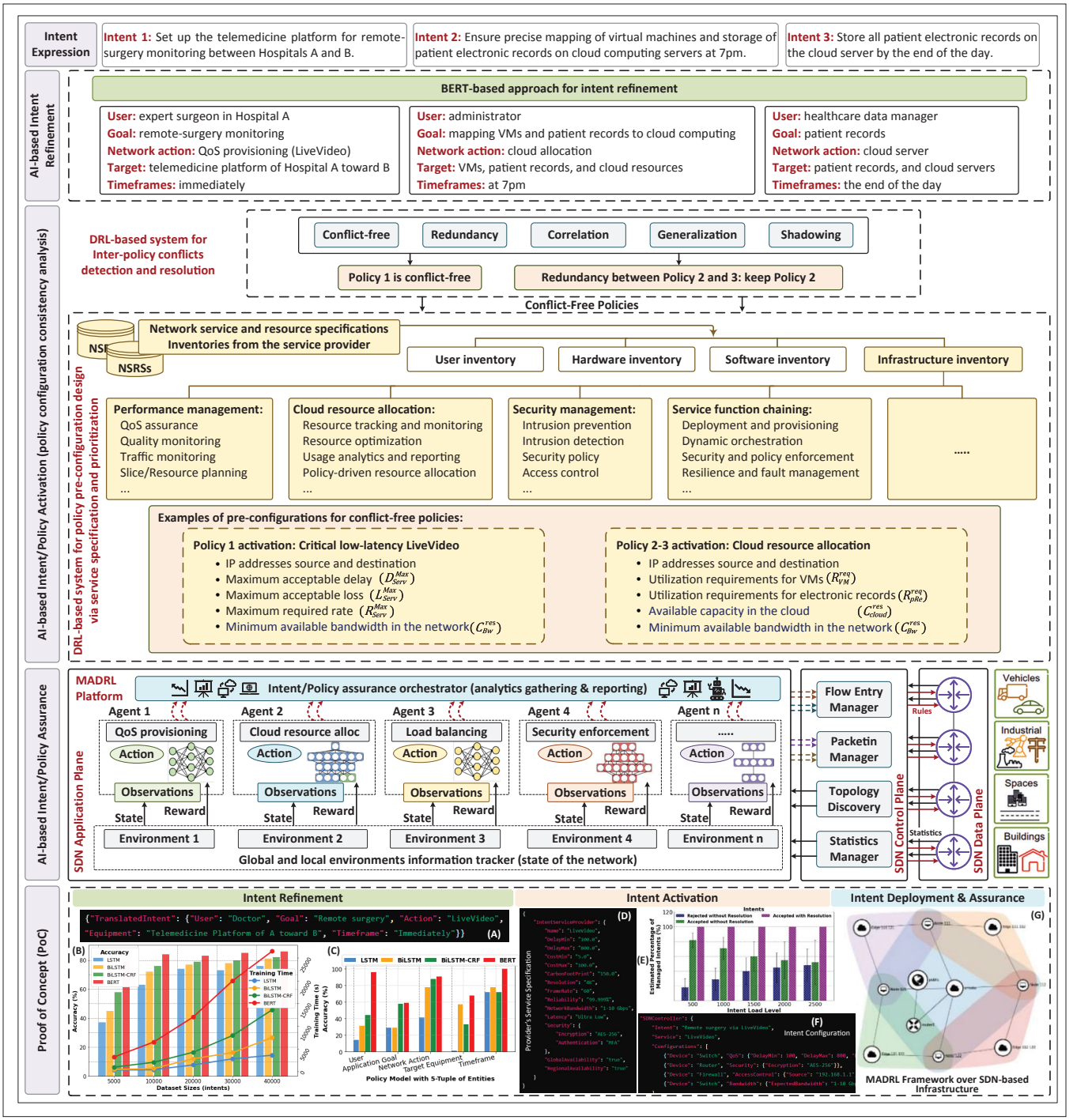


FIGURE 5. End-to-end IBNA with a Healthcare 4.0 scenario.

global objectives, ensuring intent assurance and encouraging implicit cooperation over aggressive competition for resources consumption. Various distributed algorithms, such as the independent deep Q-learning, QMix, and the deep deterministic policy gradient (DDPG) approaches, can be deployed within the MADRL platform.

Additionally, the Intent Assurance Analytics and Reporting Orchestrator oversees the MADRL agents, providing real-time analytics and coordinating dynamic policy adjustments. This enhances intent assurance, enables continuous performance monitoring, and facilitates real-time decision-making to maintain optimal network performance.

Proof of Concept

IBN is expected to be highly business specific, since the vocabulary used, along with the available infrastructure can be quite heterogeneous. To this end, we use a Healthcare 4.0 scenario, as a proof of concept, depicted in Fig. 5. In such scenario, individuals such as network administrators, data managers, hospital employees, and patients can input their requests in a natural language form.

First, the expressed intents are refined using BERT. To demonstrate the efficiency of this LLM, we compare BERT's performance against typical algorithms used for intent refinement, such as LSTM, BiLSTM, and BiLSTM-CRF, across a range

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.

The policy activation module resolves conflicts between concurrent intents (e.g., redundancy between the administrator and healthcare data manager) and validates them against service specifications before final configuration.

of dataset sizes: 5 000 – 10 000 – 20 000 – 30 000 – 40 000 user intents. The visualization in Fig. 5B aids in understanding the trade-offs between dataset size, training time, and model accuracy. The results reveal that although BERT has a higher execution time due to its computational complexity, it significantly outperforms well-known benchmarks with only 5 000 intents.

In general, the longer the dataset the easier the intent translation becomes. However, a current challenge in IBN is the difficulty of finding extensive and accurate datasets. Accordingly, with 5 000 entries, we examine in Fig. 5C the precision of correctly classifying the intent entities. BERT better identifies all five entities of the tuple, even in a low-size dataset, corroborating its superior performance against other baselines. Due to space limitations, only the precision is presented but similar observations were drawn for the other evaluation metrics. On the other hand, the evaluation of intent activation and assurance provides a performance estimation given the ongoing development of these subsystems and the lack of empirical data.

The policy activation module resolves conflicts between concurrent intents (e.g., redundancy between the administrator and healthcare data manager) and validates them against service specifications before final configuration. For instance, the refined remote surgery intent in Fig. 5A for LiveVideo activation, upon validation as conflict-free with other concurrent intents and service specifications in Fig. 5D, is subsequently activated through its final configuration in Fig. 5F. In more details, the depicted intent activation scenarios in Fig. 5E examine five different intent load levels — 500, 1 000, 1 500, 2 000, and 2 500 — both with and without a resolution mechanism. It is apparent that the percentage of managed intents for "Accepted with Resolution" remains consistently high at 100 percent, ensuring full acceptance regardless of the intent load by effectively resolving conflicts, reporting, and generating alternate policies to satisfy all intents. In contrast, the percentage of "Accepted without Resolution" shows a declining acceptance rate as the load increases, while the percentage of "Rejected without Resolution" rises correspondingly. The combined total of these two categories reflects the overall intent load level. The inclusion of error rates in the graph further enhances the understanding of system performance, particularly in the context of ongoing development.

Intent assurance, illustrated in Fig. 5G, necessitates an SDN-based environment for deploying and orchestrating heterogeneous network functionalities over the independent MADRL framework. Our ongoing investigation reveals that within this setup, RL agents autonomously learn stable individual policies, ensuring optimal network functionality. Agents are trained via environmental feedback, iteratively adjusting their policies based on received rewards and penalties. Each agent focuses on its specific task, converging independently with corresponding reward mechanisms and treating the actions of other agents as part of the environment. While agents do not need to converge together, their actions indirectly influence each other through the shared network environment. For instance, in deploying the VideoLive service, agents for QoS provisioning, traffic prediction, and security enforcement are engaged to ensure the performance. Each agent operates independently, converging on its specific functionality to collectively provide a seamless and secure video streaming experience. The performance analysis of this subsystem will be detailed in our future work.

Research Challenges and Future Directions

This section discusses the key research challenges in IBN and proposes potential future directions to address them.

Intent Refinement

Diverse stakeholder viewpoints hinder a formal intent definition creating interpretation challenges. Managing NLP vocabularies is complicated by model inaccuracies or "hallucinations," necessitating AI technique advancements like data augmentation and adversarial training. Iterative refinement and clearer intent specifications are crucial for accurate translation.

Intent Activation

Detecting and resolving intent conflicts is challenging due to platform dependency, complex service configurations, as well as due to the involvement of multiple service providers and interactions among intent handlers. An advanced and generic AI framework for dynamic policy adjustment, can help ensure consistent and conflict-free intent activation and standardize intent negotiation.

Intent Assurance

Ensuring reliable intent assurance remains a challenge due to network complexity and heterogeneity. It necessitates adaptable, automated AI models to proactively detect and resolve network functionality issues without user intervention. Further research is essential to enhance these models and facilitate knowledge sharing across different systems.

Multi-Domain IBN

Deploying intents across multiple domains presents coordination, cross-domain visibility and automation challenges. Collaboration and communication complexities across different network domains emphasize the need for synchronized controllers to standardize communication and enhance performance.

Zero-Touch Networks

Achieving fully autonomous zero-touch networks remains a significant challenge. While advanced AI models can replicate some human decision-making processes, further development of softwarization and AI models is necessary to create self-governed networks.

Conclusion

This article presented an autonomous IBNA leveraging recent trends of AI. We first reviewed the main components of an IBNA, such as intent refinement, activation, and assurance. We then proposed an AI-driven approach for each of the IBN components, integrating them into a unified pipeline. The presented pipeline not only illustrated the feasibility of deploying an AI-powered

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.

IBNA but also underscored its potential to realize a zero-touch network infrastructure.

References

[1] C. Li *et al.*, "RFC 9316: Intent Classification," 2022.
[2] S. M. Ruiyue Xu *et al.*, "3GPP: Intent Driven Management Services for Mobile NetWorks," 2023.
[3] ONF-Open Networking Foundation, Intent Framework; available: https://wiki.onosproject.org/display/ONOS/Intent+-Framework; accessed: June 3, 2024.
[4] CAMARA, Revolutionizing Telecommunications With Intent-Based Quality on Demand APIs, www.cablelabs.com/blog/revolutionizing-telecommunications-withintent-based-quality-on-demand-apis/; . accessed: June 3, 2024.
[5] A. Leivadeas and M. Falkner, "A Survey on Intent-Based Networking," *IEEE Commun. Surveys & Tutorials*, vol. 25, no. 1, 2023, pp. 625–55.
[6] A. Yichiet *et al.*, "Intent-Based Network Policy to Solution Architecting Recommendations," *Int'l. J. of Business Data Communications and Networking*, vol. 17, no. 1, 2021, pp. 55–74.
[7] Y. Ouyang *et al.*, "A Brief Survey and Implementation on Refinement for Intent-Driven Networking," *IEEE Network*, vol. 35, no. 6, 2021, pp. 75–83.
[8] D. Comer and A. Rastegatnia, "OSDF: An Intent-Based Software Defined Network Programming Framework," *Proc. 2018 IEEE 43rd Conf. Local Computer Networks*, 2018, pp. 527–35.
[9] X. Zheng, A. Leivadeas, and M. Falkner, "Intent Based Networking Management With Conflict Detection and Policy Resolution in an Enterprise Network," *Computer Networks*, vol. 219, 2022, p. 109457.
[10] C. Prakash *et al.*, "PGA: Using Graphs to Express and Automatically Reconcile Network Policies," *Proc. 2015 ACM Conf. Special Interest Group on Data Communication*, ser. SIGCOMM '15, Association for Computing Machinery, 2015, p. 29–42.
[11] A. Singh, G. S. Aujla, and R. S. Bali, "Intent-Based Network for Data Dissemination in Software-Defined Vehicular Edge Computing," *IEEE Trans. Intelligent Transportation Systems*, vol. 22, no. 8, 2021, pp. 5310–18.
[12] T. A. Khan *et al.*, "Intent-Based Networking Approach for Service Route and Qos Control on Koren SDI," *Proc. 2021 IEEE 7th Int'l. Conf. Network Softwarization*, 2021, pp. 24–30.
[13] C. Yang *et al.*, "Smart Intent-Driven Network Management," *IEEE Commun. Mag.*, vol. 61, no. 1, 2023, pp. 106–12.
[14] S. S. Mwanje *et al.*, "Intent-Driven Network and Service Management: Definitions, Modeling and Implementation," Authorea Preprints, 2023.
[15] Y. Njah *et al.*, "Toward Intent-Based Network Automation for Smart Environments: A Healthcare 4.0 Use Case," *IEEE Access*, vol. 11, 2023, pp. 136,565–76.

Biographies

YOSRA NJAH is a Postdoctoral Fellow at ÉTS, Montreal, Canada. Her current research interests include IBN, SDN, network management, performance optimization, data analytics, with emphasis on IoE and data center networks.

ARIS LEIVADEAS [SM] is an Associate Professor with the department of Software and IT Engineering at ÉTS, Montreal, Canada. His current research interests include NFV, IBN, cloud and edge computing, IoT, and network optimization.

MATTHIAS FALKNER is a Distinguished Architect at Cisco, where he focuses on enterprise and SP network architectures. His current research focuses on IBN, 5G, virtualization, and traffic modeling.

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 07,2025 at 04:57:07 UTC from IEEE Xplore. Restrictions apply.